

07/03/2022 (F.N.)

Code No: 153CF

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech II Year I Semester Examinations, March - 2022

DISCRETE MATHEMATICS

(Common to CSE(AIME), CSE(DS), CSE(IOT))

Time: 3 Hours

Max. Marks: 75

Answer any five questions

All questions carry equal marks

1. Obtain the PDNF and PCNF of the following statement formula: $(\neg P \vee \neg Q) \rightarrow (P \leftrightarrow Q)$.

[15]

- 2.a) Show that $(\exists x)(p(x) \wedge (Q(x) \rightarrow (\exists x)(p(x) \wedge \neg(x)Q(x)))$.

- b) Symbolize the following argument and check for its validity:

All men are giants.

Not all birds can fly.

Some babies are illogical.

There is a student who likes maths, but not history.

[7+8]

- 3.a) Use a Venn diagram to illustrate the relationships $A \subseteq B$ and $A \subseteq C$.

- b) Prove that the relation "congruence mod m " given by

$R = \{ \langle x, y \rangle / x - y \text{ is divisible by } m \}$ over

the set of positive integers is an equivalence relation.

[6+9]

- 4.a) Give the definition and examples of reflexive symmetric, anti symmetric and transitive relations.

- b) What is meant by Poset? Draw the Hasse diagram for the Poset.

$\langle \{2,4,5,10,12,20,25\}, \mid \rangle$ and compute LUB and GLB.

[7+8]

- 5.a) Conjecture a formula for the sum of the first n positive odd integers. Then prove your Conjecture using mathematical induction.

- b) Express the linear search algorithm as a recursive procedure and explain.

[8+7]

- 6.a) Prove that 2 divides $n^2 + n$ whenever n is a positive integer, using mathematical induction.

- b) Give a recursive algorithm to compute the factorial of a number and explain.

[8+7]

7. Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$ where $a_0 = 10$ and $a_1 = 41$.

[15]

- 8.a) Find the degree sequence of K_4 graph.

- b) What are the various ways of representing graphs? Explain them briefly with relevant examples.

[6+9]

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